



Amplification of Superpositional Effects Through Electronic–Conformational Interactions

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Abstract—Electronic–conformational interactions allow for a speedup of biomolecular structure formation that derives from the quantum mechanical superposition principle. Changing phase relationships among electronic states of different energy lead to perturbative interactions between non Born–Oppenheimer electrons and atomic nuclei that can have a self-amplifying (positive feedback) character. Conformations that satisfy the conditions for such an interaction are inherently unstable. The term “conformational soliton” refers to nuclear configurational changes that ripple through macromolecular lattices in a manner that preserves informational integrity. The model suggests that such states of ordered conformational motions are generic to polymacromolecular organizations of the type that occur in biological cells.

1. INTRODUCTION

Conformational changes appear to accompany all biomolecular recognition-action processes. An enzyme, for example, may be pictured metaphorically as interacting with a substrate in a lock–key manner; if the shapes fit together a catalytic action occurs. In reality, the shape of the enzyme changes during the course of the reaction and the response is enabled and modulated by a variety of milieu features (including control molecules). The formation of the complex is typically attributed to diffusive motion (“Brownian search”) and shape fitting augmented by conformational motion. Recent years have also seen efforts to develop a variety of dynamic pictures in which nonlinear interactions, such as electronic–conformational coupling, play a self-organizing role [1–7].

In previous publications the author has argued that the rates of complex formation and decomposition (the docking and undocking of the interacting molecules) depend on interactions between electronic and nuclear degrees of freedom that are inherently unstable [8–10]. The purpose of the present paper is to show how such interactions can draw on the parallelism inherent in the quantum mechanical wave function to generate conformational changes that propagate over long distances in macromolecular networks in biological cells. Cytoskeleton would be a prominent example; DNA strands and the bilayer membrane itself are other possible media. The term “conformational soliton” is intended to evoke the picture of an electronic–conformational instability that perpetuates itself over substantial distances in a macromolecular network in a way that requires no net dissipation. The term “self-perpetuating conformational change” could also be used, since the propagation processes resulting from electronic–conformational interactions are in many respects quite different from those of conventional solitons.

2. QUALITATIVE PREVIEW

2.1. *Dynamic free energy surfaces*

Let us begin with an informal description of the physical principles underlying docking speed. As noted above diffusion and dynamic shape fitting play a role. The probability that a "molecular lock" and a "molecular key" would find each other through random collisions should decrease as the specificity of the fit increases. To achieve both specificity and speed it is necessary that the molecules dynamically draw each other into the required relative orientation. The conformational changes that accompany recognition reflect this requirement. The greater the "drawing in" effect the lower the free energy of the complex. The likelihood of decomposition due to purely thermal agitation must then decrease. This implies that specificity and speed are conflicting requirements in a purely thermal model of enzyme catalysis.

Enzymatic reactions are nevertheless both quite specific and quite fast. This can be achieved if the complex is inherently unstable [3, 11]. The thermodynamic picture is that of a dynamic free energy surface (Fig. 1). The substrate molecule bounces into the enzyme, undergoes a catalytic modification, and then bounces out. A bouncing ball provides a simple example of such a nonthermal motion. The elasticity of the ball allows an exchange between potential and kinetic energy. In the model to be presented the exchange occurs between the electronic and nuclear coordinates. The nuclear system transfers (perturbation) energy to the electronic system, which in turn transfers it back to the nuclear system. In the presence of suitable nonlinearities (with a positive feedback character) the effect is self-amplifying.

2.2. *Quantum speedup principle*

The quantum mechanical superposition principle plays the key role in this exchange. The wave function of the interacting molecules constitutes a set of possible states that interfere with each other. Let us divide the entire system—for example, enzyme and substrate—into electrons, atomic nuclei, and hydrogen bonds (which may either be shared or mobile protons). The de Broglie wavelength of the atomic nuclei is too small for wave properties

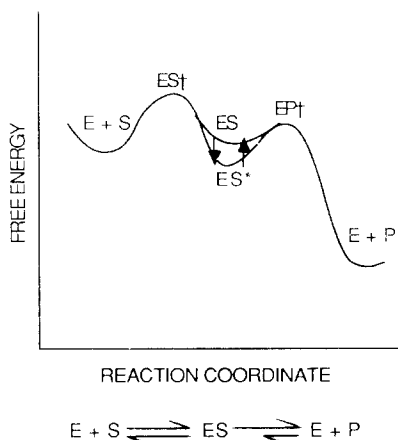


Fig. 1. Unstable free energy surface. E represents enzyme and S represents substrate. $ES^\ddagger$ and $EP^\ddagger$ are activated states. ES represents the enzyme-substrate complex. ES^* , which is not explicitly represented in the reaction scheme, is a second form of the complex that has lower energy than ES. The double potential surface indicates that ES and ES^* are connected by a nonthermal motion (e.g., an electronic-conformational interaction).

to play an important role. The possible positions of the nuclei converge to a definite value, and consequently the superposition of states is of negligible significance. The “shape state” of the complex is defined in terms of the coordinates of the nuclei, but even if we considered individual nuclei the superposition of states would be of only minor significance. The important point is that definite shape is alien to the superposition principle. Consequently quantum features cannot speed up the docking process if the configuration of the nuclei is the only factor considered, even if dynamic changes in the configuration are allowed.

The situation is entirely different for the electrons, and to some extent for the hydrogen bonds. The possible positions of an electron can differ significantly. When a measurement is made, the electron jumps into one of its states with some probability. The possible states interfere with one another, but the probability of finding the electron in any particular state is independent of time as long as the states are all of the same energy [12]. This corresponds to a stationary state. When states of different energy contribute significantly to the superposition, the probability of finding the electron in any particular state changes, corresponding to time development. The change in probability is due to constructive and destructive interference among the possible states of differing energy (the phase relationships among the different possible states would stay the same if they all had the same energy). The nuclei will be agitated by the resulting variations in the electronic charge distribution. In the model to be developed this agitation contributes to the speed of complex formation. In the presence of suitable nonlinearities the electronic states that contribute to the superposition are further spread out in energy by the motions they induce in the nuclei, leading to a self-amplifying effect that allows the nuclear configuration to jump out of local minima. This speeds the discovery of a unique minimum (or family of minima) in the case of molecular self-assembly and speeds both the formation and decomposition of the complex in the case of enzyme catalysis. Conformational solitons result when the conformation changes triggered in one macromolecule lead to unstable conformational interactions with a neighboring macromolecule.

2.3. *Breakdown of Born–Oppenheimer*

At first glance the above scenario is incompatible with the fact that the energy differences between different electronic states in biological macromolecules appear to be too great to allow for a speedup effect of the type described. Clearly self-assembly and enzyme catalysis occur in the dark. Thus in order for superpositional speedup to occur it would be necessary for the electrons of biological macromolecules to undergo transitions in response to the infrared radiation field. This appears to contradict the fact that the infrared spectra are ordinarily attributed to the vibration and rotation of the atomic nuclei. This is probably why models of the type described have not previously been seriously considered.

The above objection rests on the Born–Oppenheimer approximation, that is, on the assumption that the electrons are tightly enough bound to the atomic nuclei so that the potential due to their charge can be wrapped into the nuclear potential. This is not true for delocalized electrons. Such electrons will experience some motion relative to the nuclei, hence some accelerations that are not rigidly tied to the thermally induced accelerations of the nuclei. Accordingly, such electrons should make a contribution to the infrared radiation normally attributed to the thermal motions of the nuclei that deviates from the contribution they would make if Born–Oppenheimer were strictly accurate. Since the electrons and nuclei form a bound system the electronic levels must at the same time be discrete. Consequently closely spaced transitions must in fact exist. This would be of no significance in a system in which the nuclei are undergoing a periodic motion. Complex formation and

decomposition is a time dependent process that is not periodic, however. The infrared spectra of a complex should change in the course of its self-assembly, since degrees of freedom become constrained in different ways as the complex forms. If the Born–Oppenheimer assumption fails, some component of this change will then be due to closely spaced electronic transitions.

There is now substantial evidence that electrons can tunnel over significant distances in proteins, for example by tunneling through hydrogen bonds [13]. Surface electrons and some highly delocalized bonds or highly polarizable electrons (such as the electrons in the polar head groups of lipids) are probably also attached to the atomic nuclei in a sufficiently loose manner for Born–Oppenheimer to become an inadequate approximation. Parallel spin electrons that ordinarily are well approximated by Born–Oppenheimer could conceivably also be transiently delocalized if the nuclei to which they are attached are forced, in the process of complex formation, to be much closer together than their unconstrained equilibrium separation. The confinement of such electrons would serve to correlate their motions during short periods of downward fluctuation in thermal energy (over longer periods spin flipping through the spin-orbit interaction would be the more likely fate). During these short periods the electrons will behave as a pair with an energy that is lower than the unpaired ground state energy [2, 3, 11]. In this way states of different energy would be available; additionally the correlated motion of the pair would exert a perturbative effect on nuclei.

A more general argument, based on the principle of microscopic reversibility, implies that even mildly delocalized electrons should undergo transitions to different energy states in the course of complex formation. When the nuclear conformation changes, the potential to which electrons are subjected changes; the energy levels of the electrons thus change in the course of complex formation [14]. It might at first be thought that this is consistent with Born–Oppenheimer, since the change in the electronic levels follows the nuclear motion. However, according to the principle of microscopic reversibility the time reversed process should also be possible. In this case the changes in the electronic levels would lead the nuclear motions, and would therefore influence them. In short, if it is possible to follow it is possible to lead. (The effect of the electrons on the nuclei should not be analogized to the effect of a football on the earth, since the electromagnetic forces exerted by the two systems are equal and independent of mass. The difference between the masses of the two systems only controls the speed with which the follower system follows.)

2.4. *Quantum versus thermal fluctuations*

A second objection also suggests itself, namely that the quantum effect would be small as compared to the thermal effect. This would be true on a static potential surface. For a dynamic surface, such as the one schematically pictured in Fig. 1, the effects of thermal and quantum fluctuations are incommensurable. If the electronic–conformational interaction opens up a new pathway on the surface the controlling factor will be the barrier presented by this pathway; the issue is not whether a quantum effect is more effective than thermal agitation for jumping over (or tunneling through) a given barrier at room temperature, but whether it is more effective in allowing for reduction or sidestepping of the barrier. Here is where the nonlinearity of the interaction between the electronic and nuclear wave functions plays the crucial role. The quantum speedup effect outlined above is in a sense a thermal effect, since thermal (infrared) agitation initially spreads out the energies of the possible electronic states. The nonlinear interaction between nuclear conformation and the charge variations which result from the interference of possible

electronic states of different energy amplifies this initial thermal influence, a feature that is uniquely dependent on the superposition principle.

3. FORMAL REPRESENTATION

3.1. Superposition principle

The time development of the whole system (complex plus heat bath) is governed by some true time independent potential $V(x)$. Here x schematically denotes coordinates of all relevant particles—electrons, nuclei, and particles of the heat bath. This potential is unknown, and is in any event useless since it provides too much information about the time development of the system (we are not interested in the heat bath). At each stage of development we can suppose that some features of this true potential become relevant. We can denote this time dependent potential by $V(x_e, x_n, t)$, where x_e denotes the coordinates of the electrons of the interacting macromolecules and x_n denotes the coordinates of their nuclei. The time evolution of the potential function means that some information about it is absorbed into the heat bath and represented in terms of a statistical description.

Let us denote the electronic wave function of the complex by $\Psi_e(x_e, t)$, and the nuclear wave function by $\Psi_n(x_n, t)$. For the purposes of eliciting the speedup and instability inherent in electronic–conformational interactions it is sufficient to regard these as notational wave functions—that is, we need not here attempt to represent the detailed character of the electronic or configurational states. Also, we will work with wave functions rather than density matrices; the latter would make it possible to include thermal statistical effects more elegantly, but at the expense of losing focus on the quantum statistical effect that is here to be proposed.

If the potential energies (or Hamiltonians) of the electronic and nuclear systems were additive we could write

$$\Psi(x_e, x_n, t) = \Psi_e(x_e, t)\Psi_n(x_n, t). \quad (1)$$

This is not possible in a system in which the potentials share coordinate dependencies. In ordinary atomic or molecular problems the nuclear wave function, expressed in center of mass coordinates, is considered constant; hence the electronic levels can be static and governed by a fixed potential that is determined by the interaction between electrons and nuclei and between electrons and electrons. This is equivalent to assuming the complete absence of an electronic–conformational interaction. In our case the latter interaction is the main point of interest, and consequently the clean splitting described by equation (1) cannot be strictly accurate since the electronic and nuclear wave functions must mix to some extent.

The time scales of electronic and nuclear transitions are very different, however. Our idealization assumption is that the difference is so great that we can treat the wave functions as being sufficiently separable so we can make a quasi-independence assumption. When the molecules come into contact the wave functions are modified by an interaction term that requires them to be updated. Updating continues until the wave functions converge to mutual consistency (in which case the complex stabilizes) or until they diverge (in which case the complex breaks up). Quasi-independence means that the updating can be done independently—that is, the electronic wave function can evolve for a while, then the nuclear wave function changes, then the electronic wave function evolves again, and so forth. Apart from the updating points the potential function can then with reasonable approximation be split into two parts

$$V(x_e, x_n, t) = V_n(x_e, x_n, t) + V_e(x_e, x_n, t), \quad (2)$$

where V_e denotes the potential controlling the behavior of the electronic system and V_n denotes the potential controlling the behavior of the nuclear system. The reversal of subscripts relative to those of the corresponding wave functions (e.g. V_n is the potential for Ψ_n) is intended to indicate that the positions of nuclei make a more significant contribution to the potential governing the time evolution of the electronic system than do interactions among the electrons and that the distribution of electrons makes a contribution to the potential governing the time evolution of the nuclear configuration that is not entirely attributable to the strong and weak bonding interactions among the nuclei (taken in the sense of the interactions used in a molecular dynamics calculation). The idealization is that the potentials do not significantly depend on the same coordinates between the updates. All the nonlinearity is thus concentrated at the updating points; between these points the time development is linearizable. These are extreme simplifications. If we can elicit a significant electronic-conformational effect under these conditions it is reasonable to assume that the effect would even be stronger if the wave functions are less separable and the nonlinearity operative at all times.

For simplicity suppose that the complex has two accessible families of nuclear configurations, labeled by I and II, and two accessible families of electronic states, also labeled by I and II. Suppose that these families (or conformations) are both separated by potential barriers, at least initially. Also for simplicity we will think in terms of two wave functions, one for all the electrons of the interacting macromolecules and one for all the nuclei. In reality, of course, the potential surface would be much more complicated and the wave functions of each of the interacting molecules would ordinarily be treated as independent of one another to a good approximation; but for the purposes of exhibiting the speedup effect in a formal way it is the interaction between electronic and nuclear wave functions that is the main point of interest.

The electronic wave function can then be expressed as a superposition of the form

$$\Psi_e(x_e, t) = \sum_{i \in I} a_i(t) e_i(x_e) \exp(-iE_{e(i)}t/\hbar) + \sum_{j \in II} a_j(t) e_j(x_e) \exp(-iE_{e(j)}t/\hbar) \quad (2)$$

where the e_i are eigenfunctions of the electronic system that are most naturally associated with nuclear conformation I, the e_j are eigenfunctions most naturally associated with nuclear conformation II, the a_i and a_j are expansion coefficients, and we assume that the wave function can be split into time independent and time dependent parts. The $E_{e(i)}$ and the $E_{e(j)}$ are the electronic energies associated with the eigenfunctions e_i and e_j .

Any complete set of orthonormal basis functions could be used to express $\Psi_e(x_e, t)$. The choice of eigenfunctions allows for the most natural association between individual basis functions and particular conformations. Since the potential is time dependent it is necessary to choose the set of eigenfunctions at some particular point in time and in a way that allows for the associations noted above. A natural choice is the initial point in time, assuming the true potential is operative. Alternatively, we can regard the electronic eigenfunctions as defined over the initial potential $V_n(x_n, x_n; t=0)$ and the nuclear eigenfunctions as defined over the initial potential $V_e(x_e, x_n; t=0)$. In the present case we are considering a coupled double well potential. Consequently the electronic eigenfunctions are determined by the double well potential representing the two possible nuclear conformations, and similarly the nuclear eigenfunctions would be determined by the double well representing the two possible families of electronic states. These are of course purely conceptual definitions. It would never be possible to actually ascertain the mathematical form of these eigenfunctions starting with the double surfaces. But in principle the eigenfunctions exist and this is sufficient for the present purposes.

Some of our “notational eigenfunctions” might not be assignable in an unambiguous way to either conformation I or conformation II. This class of eigenfunctions would contribute to speedup, due to the fact that they extend over a larger region of the potential surface. If this class is ignored we can think of the superposition in a more practical way as built up from eigenfunctions calculated in the individual wells. For the purpose of demonstrating the speedup effect we can ignore the ambiguous eigenfunctions, since their presence would in fact enhance the effect.

The wave function for the collection of nuclei may, in accordance with the above conventions, be expressed as

$$\Psi_n(x_n, t) = \sum_{r \in I} b_r(t) n_r(x_n) \exp(-iE_{n(r)}t/\hbar) + \sum_{s \in II} b_s(t) n_s(x_n) \exp(-iE_{n(s)}t/\hbar). \quad (4)$$

The n_r are the eigenfunctions of the nuclear system when it is in conformation I, the n_s are the eigenfunctions when it is in conformation II, and the $E_{n(r)}$ and $E_{n(s)}$ are the energies associated with these eigenfunctions. However, the nuclear system, because of its large mass, should with high probability be assigned to a definite state. Thus if it is initially in conformation I all the b_s can for all practical purposes be set to zero. To make matters as unfavorable as possible to the speedup effect we can also assume that the nuclear configuration is initially in a single eigenstate, so that all the b_r apart from one can initially be set to zero. Practically speaking this is reasonable since the average values associated with the different n_r should peak very sharply due to the large mass.

For the electrons it is not possible to ignore superposition effects. An electron, because of its small mass, might with reasonable probability be found in a location more naturally associated with II than with I even if the nuclei are in conformation I. To make matters as unfavorable to the speedup effect as possible, let us assume that initially all the a_s are negligible and that the electronic system is initially in a single eigenstate, so that only one of the a_r is initially nonnegligible.

If some of the electrons are not tightly bound to the nuclei it will be possible for perturbations, arising for example from the radiation field, to induce transitions to eigenfunctions with energies close to that of the energy of the initial eigenfunction. The probability distribution of electronic coordinates will then be given by

$$p(x_e, t) = \sum_r a_r^*(t) a_r(t) e_r^*(x_e) e_r(x_e) + \sum_r \sum_{r'} a_r^*(t) a_{r'}(t) e_r^*(x_e) e_{r'}(x_e) \exp\left(\frac{(-iE_{e(r)} - iE_{e(r')})t}{\hbar}\right), \quad (5)$$

where $r \neq r'$. This explicitly demonstrates the statement made earlier that the electronic charge density will change with time due to interference among the eigenfunctions of different energies, since these are associated with different frequencies of vibration. Our simplifying assumption that the wave function can be split into time independent and time dependent parts serves to elicit this well known quantum mechanical feature in an explicit manner; but in fact the assumption is more restrictive than necessary.

Equation (5) can be used to give more precise meaning to the phrase “parallelism of the quantum mechanical wave function”. The structure of the wave function is determined by two factors: the choice of the basis functions and the superposition of these basis functions. The contribution of these two factors to the “possibility space” of a system cannot be separated. Even a single basis function, say an eigenfunction, will allow for penetration into the classically forbidden region of a barrier. It might therefore appear that the quantum speedup effect could occur without interference between states of different energy. This is not the case, however, since interference inevitably occurs if the time

evolution of the system is considered. Suppose, for example, that an electron initially has a high amplitude to be in a particular potential well. With time the amplitude for being in a second well increases. The energy of the electron at the later time will be smaller, since the space in which it can reasonably be found will be larger. (This can be seen from the position-momentum uncertainty principle. An increase in the spatial range of the electron allows for a decrease in the uncertainty of its momentum, and therefore for a decrease in the kinetic energy fluctuations it exhibits.) Consequently the time evolution will involve interference among states of different energy regardless of the initial choice of basis functions. The treatment that follows will emphasize superposition, since interference is definitive of time evolution. However, the intimate connection between the choice of basis functions and their superposition must be kept in mind, since the interference pattern, and all physical properties, are independent of any acceptable choice.

3.2. *Self-consistent field*

The electronic and nuclear wave functions (equations 3 and 4) must be consistent with each other. This suggests a Hartree-type self-consistent field approach. Ordinarily this is viewed as an approximation procedure. For example, the electronic wave function of a multi-electron atom is approximated by starting with an initial trial wave function and potential and iterating until convergence is achieved. The steps toward convergence are ordinarily not viewed as representing the self-organization of the multi-electron atom, and accordingly the trial wave function need not represent any physical reality; questions about how such atoms organize themselves are not ordinarily considered. The objective is to obtain a description of a final, static system of electrons which presumably forms so rapidly and in a way so controlled by energy that questions about the formation process are not of major phenomenological interest. The situation is completely different for the formation of macromolecular complexes, such as the enzyme-substrate complex. The self-organization and self-disorganization of the complex, and the specificity of these processes, are the phenomenological features of greatest importance.

The idea, then, is to use a Hartree-like self-consistency scheme to represent, in a conceptual way, the evolution of the complex to a self-consistent state (which may ultimately be the dissociated state if no self-consistent complex is possible). Could it be that inconsistencies are possible in nature? The answer is certainly yes. A room controlled by a thermostat exhibits an inconsistency between the state of the room (ambient temperature) and the state of the thermostat (curvature of a piece of metal). If the relation between these two states is one of negative feedback the ambient temperature converges to a constant value, with some slight oscillation about this value. If the relation is one of positive feedback the temperature diverges. From the point of view of nonlinear dynamics the state of the room and the state of the thermostat are of course inconsistent in a consistent way. Our hypothesis is that the state of the electronic and nuclear systems can be inconsistent in just this fashion, that is, can be in a relation of either positive or negative feedback.

From the assumption (equation 2) that the potentials can be split in an additive manner we can express the electronic and nuclear wave functions as solutions of the Schrödinger equations

$$-\sum_i \frac{\hbar^2}{2m_i} \nabla_i^2 \Psi_e(x_e, t) + V_e(x_e, x_n, t) \Psi_e(x_e, t) = i\hbar \frac{\partial \Psi_e(x_e, t)}{\partial t} \quad (6a)$$

$$-\sum_i \frac{\hbar^2}{2M_i} \nabla_i^2 \Psi_n(x_n, t) + V_n(x_e, x_n, t) \Psi_n(x_n, t) = i\hbar \frac{\partial \Psi_n(x_n, t)}{\partial t} \quad (6b)$$

Here m_e is the mass of the electron, ∇_j^2 is the Laplacian operator of the j th electron, M_k is the mass of the k th nucleus, and ∇_k^2 is the Laplacian operator of the k th nucleus.

The complex must form from a large fraction of the initial conditions that correspond to initial contacts. The initial potentials governing the electronic and nuclear systems, since they are time dependent, would in fact then not be independent of the initial conditions. Let us suppose some initial arbitrary contact between the interacting molecules. We suppose that it is possible in principle to construct initial potentials corresponding to this initial arrangement. The initial electronic and nuclear wave functions are in principle given by the equations (6a) and (6b). The average positions of the electrons are then obtained from equation (5), and the average positions of the nuclei may be obtained analogously (except that interference effects should not be noticeable). The first cycle is completed by using these new average positions to update the potentials. The process is repeated until convergence is achieved (corresponding to mutually consistent electronic and nuclear wave functions) or until the system diverges to a self-consistent state. Divergence means jumping from one metastable state to another, leading eventually to a self-consistent complex. It is entirely possible that no self-consistent complex exists, apart from the dissociated state. Two interacting molecules could in this case jump, through a series of divergent processes, to a unique minimum that is suited to a particular catalytic function and then diverge to a dissociated state.

The updating of the potentials may be expressed notationally by writing

$$V_e(x_e, x_n, t) = V_e(\Psi_e(x_e, t), \Psi_n(x_n, t)) \quad (7a)$$

$$V_n(x_e, x_n, t) = V_n(\Psi_e(x_e, t), \Psi_n(x_n, t)), \quad (7b)$$

where the functional notation expresses the fact that V_e and V_n depend on features of both the electronic and nuclear wave functions. As noted above the updated functional forms utilize the probabilities

$$p(x_e, t) = \Psi_e^*(x_e, t)\Psi_e(x_e, t) \quad (8a)$$

and

$$p(x_n, t) = \Psi_n^*(x_n, t)\Psi_n(x_n, t). \quad (8b)$$

But of course some criteria would have to be used for obtaining adequate potentials from these probabilities; for example, interactions may be considered irrelevant if the likely coordinate values are such that the particles are more than some specified distance apart, or some averaging may be done beyond a certain distance. Updating the potentials in this manner does not represent any process inherent in the phenomena—in particular it does not imply that wave function reduction is occurring. New potentials could, with some less intuitive procedure, be obtained without calculating probabilities from the wave function. To construct a theoretical demonstration of the speedup effect it is only necessary to assume that relevant potentials exist in principle.

To formally express the scheme it is convenient to write the Schrödinger equations (6a) and (6b) in the abbreviated form:

$$\Psi(x, t + \tau_b) = \Psi(V(x, t + \tau_a), t + \tau_b). \quad (9)$$

Here $\Psi(x, t + \tau_b)$ is the (electronic or nuclear) wave function at time $t + \tau_b$, $t + \tau_a$ (with $\tau_b > \tau_a$) is the time at which the (nuclear or electronic) potential $V(x, t + \tau_a)$ is turned on, and $\tau_b - \tau_a$ is the time interval over which this potential remains applicable. The assumption that time dependent perturbation theory is adequate for dealing with any changes in the potential functions during the intervals means that the potentials take the

form

$$V(x, t) = V'(x) + v(x, t), \quad (10)$$

in a piecewise continuous manner, where $V'(x)$ is a time independent unperturbed potential applicable during the interval $\tau_k - \tau_n$ and $v(x, t)$ is a small time dependent perturbation that develops over this interval. Written explicitly in terms of what we have called the potentials for the electronic and nuclear subsystems

$$V_e(x_e, x_n, t) = V'_e(x_e, x_n) + v_e(x_e, x_n, t) \quad (11a)$$

$$V_n(x_e, x_n, t) = V'_n(x_e, x_n) + v_n(x_e, x_n, t). \quad (11b)$$

This assumption, as noted above, is more restrictive than necessary, but it allows us to separate the solution of the Schrödinger equations for $V'(x)$ into time independent and time dependent components

$$\Psi'(x, t) := \psi'(x)\phi(t), \quad (12)$$

in a piecewise continuous manner and therefore to represent the vibratory terms in equations (3) and (4) in an explicit manner. The $\Psi(x, t)$ can then be written in terms of $\Psi'(x, t)$ with time varying coefficients. (Note: we have here put the prime on the unperturbed potential so as to avoid carrying it through in the discussion to follow.)

Let the time interval $t_{n+1} - t_n$ denote the n th cycle of nuclear-electronic interactions. The electronic system is pictured as undergoing an evolution over some time interval τ_e and then as remaining constant while the nuclear system follows over a time interval $\tau_{n(n)}$. The subscript (n) associates these intervals with the start time of the cycle. The time evolution of the complex may be described by an iterative (recursive) process of the following form:

$$\Psi_e(x_e, t_n + \tau_{n(n)}) = \Psi_e(V_n(\Psi_e(x_e, t_n), \Psi_n(x_n, t_n)), t_n + \tau_{n(n)}) \quad (13a)$$

$$\Psi_n(x_n, t_n + \tau_{n(n)}) = \Psi_n(V_e(\Psi_e(x_e, t_n + \tau_{n(n)}), \Psi_n(x_n, t_n)), t_n + \tau_{n(n)}) \quad (13b)$$

in each interval the potential comprises a time independent part and a small time dependent perturbation, in accordance with equations (11a) or (11b). The sequence of cycles is initially entered by setting the potentials at time t_n , which subsumes initial wave function Ψ_e and Ψ_n at time t_n . These potentials are used to solve for Ψ_e at time $t_n + \tau_{n(n)}$. The time index $n = 0$ is then updated to $n = 1$, $\Psi_e(x_e, t_f)$ and $\Psi_n(x_n, t_f)$ in equation (11a) are set equal to $\Psi_e(x_e, t_n + \tau_{n(n)})$ and $\Psi_n(x_n, t_n + \tau_{n(n)})$, respectively, and the cycle thereby re-entered. The potential V_e is updated and used to solve for $\Psi_e(x_e, t_n + \tau_{n(n)})$, and so on.

The term "solving" could be replaced by "evolving in accordance with" since we are interested in a conceptual representation of the complex formation and decomposition process. A bona fide calculation scheme would have to introduce auxiliary procedures in approximating the wave functions and potentials. For example, the initial potentials and wave functions at time t_n should represent the true situation, in principle these potentials and wave functions exist, but in practice they would have to be calculated or in some way measured; this auxiliary calculation process would have no necessary relationship to the self-organizing dynamics of the molecules.

In a practical calculation specific reference to different time intervals between significant changes in either the electronic or nuclear state might be dispensed with, since there would be no way of knowing what these time intervals are in advance without some kind of look-ahead procedure (e.g. using the change in the probability distribution of the electrons as criterion). The intervals could be made arbitrarily small, so that one could simply switch back and forth between equations (6a) and (6b). However, these intervals serve to

highlight the feature that the nuclear motion is slower than the electronic motion. The electrons adjust rapidly to a change in the nuclear coordinates (in which case $\tau_{e(u)}$ is small), but they can also exert a leading effect due to the charge variations accompanying interference. A more descriptive notational picture might therefore utilize two electronic time constants—a fast one for the rapid readjustment to the change in the nuclear configuration and a slow one for the interference effect [8, 10]. In the present formalism the evolution from $\Psi_e(x_e, t_u)$ to $\Psi_e(x_e, t_u + \tau_{e(u)})$ on re-entering equation (13a) represents both rapid readjustment and the slower developing interference effect.

In reality the division into two interacting subsystems, electrons and nuclei, is a great oversimplification. The hydrogen bonds should be treated as a subsystem in their own right. As noted earlier, the de Broglie wavelength is large enough to allow for some tunneling motions. At the same time hydrogen bonds serve as preferred pathways for electron tunneling. The tunneling electrons are highly delocalized and therefore good candidates for exhibiting interference effects among possible states that are accessible through infrared perturbation. The hydrogen bonds are the nuclei that will be most agitated by the consequent variations in the charge distribution, and therefore the first nuclei to amplify the perturbation to the electronic system. As the variations in the charge distribution increase, the heavier nuclei will be increasingly affected, further contributing to the amplification effect. The possible sequences of events are much more complicated than in the two component schemes described by equations (13a, b). The scheme would obviously be even more complicated if the particular orbital characteristics of the various electrons were taken into account.

3.3. *Processing the wave function*

The iteration scheme (equations 13a, b) processes the superpositions of electronic and nuclear states (equations 3 and 4). Let us consider what can happen as these superpositions are iterated. Recall that the possible nuclear states are so closely spaced in energy that the fact of superposition can be ignored in favor of a classical description. The worst case assumption is that the electronic system is in a single eigenstate associated with nuclear conformation I. Conceivably the electronic wave function is consistent with the nuclear wave function. Conformation I will then be metastable, and could for all practical purposes be stable. For example, we can suppose that II is a lower energy state, but that the barrier between I and II is sufficiently high that thermal fluctuations are very unlikely to trigger a transition; quantum fluctuations in the static case would even be more unlikely to trigger a transition.

Suppose, however, that the perturbation to the electronic system induces transitions to other electronic states. The magnitude of the initial nonnegligible expansion coefficient will decrease, and the magnitude of one or more other expansion coefficients will increase. The nuclei will be agitated by the changing phase relationships among electronic states, since more than one state now has a nonnegligible probability amplitude. The initial perturbation may be from the radiation field, but the resulting agitation of the nuclei serves as an additional source of perturbation. This means that the expansion coefficients of the electronic subsystem will have a functional dependence on the expansion coefficients of the nuclear system, since both contribute to V_n .

Let us trace through some possible “execution sequences” of the wave functions in a more formal way. The original electronic wave function is determined, according to our maximally unfavorable assumption, by a single expansion coefficient $a_e(t_0)$. Due to infrared perturbations and variations of the potential due to thermal motions of the nuclei this is converted to a superposition with some set of nonnegligible expansion coefficients:

$$a_i(t_0) \rightarrow \{a_i(t_0 + \tau_{i(m)}), a_i(t_0 + \tau_{i(m)}), a_{i'}(t_0 + \tau_{i(m)})\}_{i \in I} \quad (14a)$$

Here $i, i', i'', \dots$ all belong to the index set of conformation I. The number of elements in this set is at least two, since by assumption one or more electrons are not closely tied to the nuclei. Interference phenomena, including charge variations, thus appear. The wave function of the nuclei is determined by an equivalence class of eigenfunctions that are hardly distinguishable from the macroscopic point of view. To keep the notation simple we will ignore this distinction and, as previously indicated, denote the nonnegligible expansion coefficient by a single expansion coefficient. Because of the electronic charge variations the initial nonnegligible equivalence class $b_i(t_0)$ undergoes the transition

$$b_i(t_0) \rightarrow b_{i'}(t_0 + \tau_{i(m)}). \quad (14b)$$

The wave function defined by the new equivalence class of coefficients $b_{i'}$ corresponds to a new configuration of the nuclei, which we suppose (so as not to beg the question) is still assignable to conformation I. The electronic system undergoes transitions to new states in response to the nuclear motions:

$$\{a_i(t_0 + \tau_{i(m)}), a_i(t_0 + \tau_{i(m)}), a_{i'}(t_0 + \tau_{i(m)})\}_{i \in I} \\ \rightarrow \{a_i(t_0 + \tau_{i(m)}), a_{i''}(t_0 + \tau_{i(m)}), a_{i'}(t_0 + \tau_{i(m)})\}_{i \in I} \quad (14c)$$

This corresponds to re-entry into the second cycle, and therefore involves a fast readjustment of the electrons to the new nuclear potential (V_n at time $t_0 + \tau_{i(m)}$) and also the slower interference effect. The latter effect now has three sources: the infrared radiation field and thermal motions of the nuclei, as before, and in addition the agitation of the nuclei that results from the interference itself. This additional source of perturbation remains after the fast readjustment unless it is assumed that all the transitions go directly to a single electronic state. In reality of course the agitation of the nuclei begins at time t_0 and acts reciprocally on the electrons on a continuous time scale. All this is represented (conceptually) in the updated V_n at $t_0 + \tau_{i(m)}$, or equivalently at time t_1 . The time $t_0 + \tau_{i(m)}$ and time $t_1 + \tau_{i'(m)}$ sets might be entirely disjoint, or there may be some overlap.

Completing the $n = 1$ cycle

$$b_i(t_0 + \tau_{i(m)}) \rightarrow b_{i'}(t_0 + \tau_{i(m)}) \quad (14d)$$

and entering the $n = 2$ cycle

$$\{a_i(t_0 + \tau_{i(m)}), a_{i''}(t_0 + \tau_{i(m)}), a_{i'}(t_0 + \tau_{i(m)})\}_{i \in I} \\ \rightarrow \{a_i(t_0 + \tau_{i(m)}), a_{i''}(t_0 + \tau_{i(m)}), a_{i'}(t_0 + \tau_{i(m)})\}_{i \in I} \quad (14e)$$

The wave function defined by the equivalence class of coefficients $b_{i'}$ might still be assignable to conformation I, but conceivably they are assignable to II. If so the $t_0 + \tau_{i(m)}$ set should also be predominated by coefficients belonging to conformation II, since the electronic system should make a fast readjustment to the new conformation. The molecule will then be locked in conformation II, assuming that this can eventually evolve to a self-consistent state or to a cycle of quasi-consistent states all of which belong to conformation II. If II does not admit such a state or set of states the complex will eventually switch back to conformation I (since the example has been unrealistically restricted to only two conformations).

Feedback between a quantum and a classical system is the important feature illustrated by the above sequence. *The superposition of possible electronic states serves as the channel through which motions of the nuclear system are fed back into the nuclear system.* If the feedback is positive the motions of the nuclei will be self-amplifying. The complex will exhibit increasingly agitated motion, and then undergo a sudden significant change of form.

If the feedback is negative the electronic and nuclear system evolve to mutual consistency. Negative here means that the charge variations resulting from interference push the nuclei back towards their initial arrangement. The nuclei will then oscillate around some equilibrium configuration. The vast majority of molecules fall to a state of lowest energy that is for all practical purposes completely consistent and time independent. From our point of view the feedback loops are present but of negligible importance in these cases. They are unmasked only in those molecules in which the nuclear motions are intricately constrained in three dimensions (conformation is important) and in which electrons are present that can move somewhat independently of the nuclei.

From equations (6a, b) and (7a, b) it can be seen that the potentials are, through the updating procedure, functionally dependent on each other. This is an alternative way of looking at the time dependence of the expansion coefficients and the coupling of these coefficients.

3.4. *Connection to fluctuation-dissipation*

Fluctuation is necessary to induce transitions of the electronic system to states of different energy. The time development of a perturbed molecule would then ordinarily involve dissipation of energy into the heat bath in accordance with the equipartition of energy. The nonlinearity of the electronic-conformational interaction allows this dissipation to be focused into selected degrees of freedom of the nuclear configuration. The self-amplifying (positive feedback) form of the interaction is driven by fluctuation in a focused way over an extended period of time. The agitation of the nuclei can serve to bounce energy back to the electronic system in a selective way, but the self-amplification would die out in the absence of heat bath fluctuation.

Fluctuation-dissipation is concomitant to all chemical reactions, since fluctuation is the source of collisions that drive reactions over activation barriers. No macroscopic work is performed. What is controlled, as in all catalytic processes, is the rate at which work is performed and the selectivity of the pathway.

4. COUPLED INSTABILITIES

So far we have pictured the complexing of two molecules, at least one of which must be a macromolecule. The formation of the complex is described by a dynamic free energy surface. The parallelism inherent in the quantum mechanical wave function (i.e. in the superposition of electronic states) speeds the search of this potential surface, and also greatly increases the likelihood of finding a unique minimum that is independent of the initial conditions. It does so by agitating the nuclei in a positive feedback fashion, thereby opening up new pathways for formation or decomposition, or lowering or narrowing existing barriers. If the complex that is formed is self-consistent, or if it allows for cycling among a family of conformationally equivalent states, we have self-assembly. The folding of an individual protein fits into this situation. The folded protein maybe viewed as self-organizing from the interactions among the different regions of a single covalently bonded sequence of amino acids. Enzyme-substrate interactions correspond to the situation of an unstable complex. If the enzyme recognizes the substrate it makes or breaks a particular covalent bond and then ejects it; if it does not recognize the substrate it dissociates without taking an action. Conformational switching in response to control molecules (allosteric control) can be interpreted in a similar manner.

Let us now turn to supermacromolecular complexes. We can think of these as highly structured arrangements of macromolecules that self-assemble in jigsaw puzzle fashion.

Such polymacromolecular constructions are ubiquitous in biology. They come in the form of extensive sheets (membranes), localized but highly intricate constructions (chloroplasts, mitochondria, ribosomes), wide ranging but highly structured networks (the cytoskeleton), and linearly oriented architectures (chromosomes). These structures exhibit motions that help to choreograph the flow of components in the cell at all levels, ranging from directed movement of electrons in chloroplasts and mitochondria to the micromuscular activities of the cytoskeleton. In all likelihood they also serve as channels for signal flows that coordinate distant events.

For simplicity we will picture these polymacromolecular constructions as lattices, with each lattice point occupied by a macromolecule. Each molecule potentially interacts with any of its immediate neighbors. The interaction can be viewed in analogy to the interaction between an enzyme and substrate, except that in this case we are not concerned with the making or breaking of a covalent bond. The "substrate" acts solely as a control molecule that switches the enzyme into a particular conformation. The macromolecules of the lattice can thus act as control molecules relative to one or more of their neighbors and can act as controlled molecules relative to one or more neighbors.

Figure 2 illustrates in schematic fashion how interactions of the type proposed in the previous section could lead to a propagating conformational change in a one-dimensional lattice built from a single type of macromolecule. Each molecule has two conformations available to it, I and II, as in our previous discussion. Conformation I is represented by a left handed arrow and conformation II by a right handed arrow. Conformation I is self-consistent (stable) except when it complexes with a conformation II molecule, in which case it undergoes a transition to conformation II (complexing is represented by two arrowheads pointing towards each other). Similarly conformation II is stable except when it complexes with conformation I, in which case it reverts to conformation I. The initial conformational switch is triggered by an externally imposed input on the far left hand side of the chain. The conformational change propagates through the chain until it reaches the right hand side, where it triggers an effector molecule into an active state.

To the above lattice we have attributed properties inherited from the superposition-amplification model described in the previous section and characteristic of the biological examples mentioned above. The subsumed conformational changes are significant, they have a high degree of directional specificity, and the rate of propagation is high. The changes can propagate over substantial distances in a homogeneous lattice in a manner that maintains their original form. In an inhomogeneous lattice the conformational change would

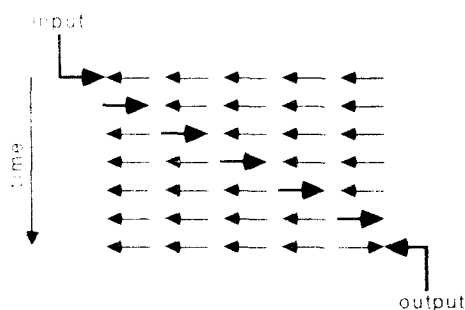


Fig. 2. Schematic representation of a conformational soliton propagating through a one-dimensional polymacromolecular lattice. The first frame (top line) represents the presence of an external input that triggers the initial switch in conformation. Each subsequent frame represents the migration of the switch to the next lattice position. The last frame (bottom line) represents the triggering of an output action.

itself change in character depending on the particularities of the neighboring molecules, but could do so without losing its informational integrity.

The above properties are incompatible from a classical point of view, but are generic to superposition–amplification models since in these the time development is generated by inherent instabilities. In a classical model purely thermal fluctuations could lead to switching between significantly different conformational states, but the potential barriers would have to be low and consequently the directional specificity with which the conformation changes ripple through the lattice could not be high. If the potential barriers were high enough to confer high selectivity on the propagation patterns, the rate of propagation would be low. In the absence of high specificity and high speed the spatial range over which it would be possible to maintain informational coherence would be compromised. But according to our superposition-amplification model the chance that a macromolecular aggregate exhibits these properties simultaneously becomes more rather than less likely as the architectural complexity of the aggregate increases, since the chance of satisfying self-consistency conditions decreases as the number of constraints that have to be satisfied increases.

5. CONCLUDING REMARKS

The new feature of our model is the manner in which the linear superposition principle is embedded in a nonlinear interaction. The nonlinear interaction between the electrons and nuclei constitutes a feedback loop, but one that cannot be treated in its entirety as if all elements are describable by classical variables. The rich structure of the quantum mechanical wave function is drawn into the loop and hence contributes to specificity and speed.

The nonlinearity is here built on top of linear quantum mechanics, not built into it in a fundamental way. It enters in the same way as it would if Newton's equations were used—by focusing attention on major components of a system and ignoring details of the heat bath. Our analysis begins with the standard Schrödinger equation. In complete microscopic detail the admissible solutions would be confined to unitary transformations (translations and rotations in Hilbert space, no stretching or folding). The time dependence of the potentials in our model and the treatment of the nuclear configuration in classical terms rests on the departure from a completely detailed description. The time development of the system of interest—the interacting molecules *sans* heat bath—is no longer confined to unitary transformations. Exotic dynamic phenomena can occur that would appear to be excluded by the Schrödinger equation.

Sensitive dependence on initial conditions suggestive of chaos is an example. Divergence of trajectories (stretching) can be physically interpreted in terms of the spreading out of the electronic wave function. The consequent interference induces an irregular motion of the nuclei. This is the phase of the electronic–conformational interaction in which the electrons lead. Confinement of trajectories to some domain (folding) can be interpreted in terms of the adjustment of the electronic levels to changes in the nuclear configuration. This is the phase of the interaction in which the electrons follow. If the number of nonnegligible eigenfunctions is very small the electronic structure will approximate time independence; if it is very large an ergodic, statistically regular regime will be approached. Between these extremes the nuclei should undergo a series of configurational changes that are highly irregular and initial condition sensitive. If folding (confinement) dominates the series will belong to a single conformation. This is the negative feedback situation. If stretching (divergence) dominates jumping to a new conformation (basin of attraction) will occur. This is the positive feedback situation.

The model makes a number of experimental predictions of a general nature. Sustained low frequency radiation should affect the rate of complex formation by inducing closely spaced transitions among possible electronic states. It should be possible to introduce color groups into proteins in such a way that folding and complex formation are disrupted in the presence of visible light, but not in the dark. It should be possible to alter the rate of complex formation by replacing hydrogen atoms by deuterium or tritium; the effect would be pronounced for those hydrogen bonds that provide preferred pathways for electron tunneling. If confinement of parallel spin electrons can augment superpositional speedup by inducing transient delocalization this should be detectable as a short-lived alteration of a spin resonance signal prior to a shift in conformation.

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